



CC106 APPLICATION DEVELOPMENT WITH EMERGING TECHNOLOGY

PROJECT PROPOSAL

Instructions: Fill in all required fields. Write brief but clear answers. This is a **course project proposal**, not a capstone paper.

A. STUDENT INFORMATION

- **Student Name(s):**
ARBOLADO, CARL V.
BONDILLES, JOHN ANDREI S.
CABILING, ALYANA DOMINIQUE G.
DE GUZMAN, MURPHY B., JR.
HANABAN, GYDRHICK EYVAN R.
RIMORIN, LOIS JAY D.
SALGADO, JULIA H.
- **Course / Section:** SBIT- 3P
- **Instructor:** Prof. Mary Jean Jayobo
- **Date Submitted:** January 29, 2026

B. PROJECT TITLE- RIMORIN

- **Proposed Project Title:** Nutrisense: An Intelligent Health and Wellness mobile Application for Students

C. PROJECT OVERVIEW- BONDILLES

1. **Brief Description of the Mobile Application (2–3 sentences):**
-NutriSense is a health-focused application designed to help students learn how to develop meaningful and practical digital solutions that promote nutrition and overall well-being. The project introduces learners to the fundamentals of designing a health application that collects, analyzes, and interprets user health data to provide personalized food and nutrition recommendations.
2. **Target Users:**
☒ Students ☐ Teachers ☐ Employees ☐ General Public ☐ Others:



3. Mobile Platform:

- ☒ Android ☐ iOS ☐ Cross-platform

D. EMERGING TECHNOLOGY TO BE USED- ARBOLADO

1. Selected Emerging Technology:

- ☒ Artificial Intelligence (AI) / Machine Learning
☐ Internet of Things (IoT)
☐ Augmented Reality (AR)
☐ Blockchain
☒ Cloud-based API / Services
☐ Others: _____

2. Brief Explanation of the Technology (2–3 sentences):

We will use Artificial Intelligence to create an effective schedule based on user input, particularly for school timetables. In addition, an API will be used to collect and gather a list of available foods, which can then be organized according to the user's preferences and the foods that are accessible.

3. How will this technology be applied in the mobile app?

By integrating the API link into our mobile application, we will be able to retrieve real-time data such as available food options and other relevant resources. The app will filter and organize this information based on the user's preferences and accessibility. At the same time, Artificial Intelligence will be applied to analyze user inputs (such as class schedules and daily routines) to automatically generate an effective and personalized timetable, helping students manage their time more efficiently.

E. PROBLEM / RATIONALE- CABILING

1. Problem or Need Addressed by the Application:

Many students experience difficulty in maintaining a healthy lifestyle due to stress in school, heavy academic workloads, and limited time. These challenges can lead to poor eating habits, skipping meals, and lack of regular physical activities. In addition, some students may not know how to prepare healthy meals using the food already available in their home or creating a workout routine that fits in their available schedule

2. Why is this application needed?

This application is needed because many students struggle in having a



healthy lifestyle due to stress, academic workloads, and having a limited time that may lead to poor eating habits, skipping meals and lack of physical activity which can affect student's physical health, mental well being and their academic performance

Students may also lack knowledge about proper nutrition or feel that eating healthy and exercising require too much time, money, or effort. As a result, they rely on fast food, skip meals, or avoid workouts altogether.

This application provides a practical solution by offering healthy meal suggestions based on available ingredients at home and creating workout plans that fit into students' daily schedules. This makes healthy living more accessible, convenient, and sustainable for students.

This application is needed because it helps students to balance stress, time constraints and lack of knowledge in maintaining proper nutrition and regular physical activity while managing their academic responsibilities.

F. OBJECTIVES OF THE PROJECT- DE GUZMAN

1. **General Objective:**

To encourage overall health and wellness by creating an intelligent system that enables people to reach their diet and fitness objectives through personalized calorie tracking, food suggestions, and fitness plans according to their preferences.

2. **Specific Objectives (check at least three):**

- ☒ Design a user-friendly mobile interface
- ☒ Implement core mobile application features
- ☐ Integrate the selected emerging technology
- ☒ Test basic functionality of the application
- ☐ Others:

G. PROPOSED SYSTEM FEATURES - SALGADO

List the main features of the application:

1. Personalized Health Profiling
2. AI-Based Scheduling Recommendations.
3. Personalized Food and Recipe Recommendation System.
4. Smart Reminder and Notification System.
5. Wellness Analytics Dashboard.



H. SCOPE AND LIMITATIONS - SALGADO

1. Scope (What the app can do):

1. The system will recommend a workout plan and schedule based on the student's class schedule.
2. The system will recommend nutritious foods depending on the items available at home and the student's mood, and medical conditions as indicated by the record.
3. The system will assign daily quests related to their health goals.
4. The system will send a reminder for hydrations, sleep, physical activity and mental breaks.

2. Limitations (What the app cannot do / constraints):

1. The system will not be integrated into any type of vital sign monitoring equipment as it is unable to track the student's vital signs in real time.
2. The user's profile cannot be shared through any social media accounts.
3. The system will only operate online mode.

I. DEVELOPMENT METHODOLOGY - HANABAN

● Development Approach:

☒ Agile ☐ Incremental ☐ Waterfall ☐ Others: _____

● Brief Description of Development Steps:

The Agile Development Approach will be used in this project to allow flexibility and continuous improvements for the project. The project will be developed in iterative cycles that allows features to be planned, designed, implemented, and tested. Each cycle will focus on improving the system based on feedback and suggestions. This approach ensures flexibility and continuous improvement throughout the development process.

J. TOOLS AND TECHNOLOGIES - ARBOLADO

- **Programming Language(s):** Dart Programming Language
- **Framework / IDE:** Flutter Framework / Android Studio or VS Code
- **API / SDK (if any):** Flutter SDK / DART SDK
- **Database (if any):** Firebase Firestore

K. EXPECTED OUTPUT

- **Working mobile application (prototype/basic version)**



- Source code
- Short documentation
- Project demo

L. PROJECT TIMELINE - RIMORIN

Week	Planned Activity
2	Topic approval - Brainstorming of project ideas - Project concept - Instructor approval
3	Detailed Proposal - Assigning of task per member - Write project backgrounds & objectives - Identify target users - Listing system features - Choose tools and technologies
4	Presentation of Project Proposal - Assigning task per member - Create presentation slides - Present timeline& features - Receive feedback
5 - 6	Data Gathering - Interview/ survey users - Research existing or similar system - Collect requirements - Finalize system flow and features
7 – 8	UI/UX design - Plan the design - Draw wireframes - Design UI layout
10 – 12	Application development



Week	Planned Activity
	- Set up project environment - Create database & tables - Code main features - Integrate UI & backend
13 - 14	Testing & debugging - Project consultation - Test each feature - Fix errors & bugs - Improve performance - Validate user inputs
15 - 16	Implementation - Prepare and deploy for actual use. - Prepare demo data - Monitor system for further improvement and initial issues
17	Final submission & demo - Final Presentation - Submit source code & documentation - Demonstrate system functionality

M. CONCLUSION (1–2 sentences)- Bondi

Briefly explain the learning value and expected outcome of the project.

- This application helps students learn how to develop and design a useful and meaningful health application that focuses on nutrition and overall well-being. Through this application, learners will be able to gain more and deeper understanding of how user health data can be collected, analyzed, and used to give personalized food recommendations based on the body's needs. The project also builds skills in logical thinking and decision making by allowing the app to suggest meals using ingredients available in their home. In the end, NutriSense aims and focuses to support users in making informed, balanced, and healthier food choices, while showing how technology can play an important role in improving daily lifestyle and long-term health.

Approved by Instructor:



MARY JEAN M. JAYOBO
Signature over printed name

Date: _____

Provide a screenshots of your group collaboration meeting. ONLINE / F2F





QUEZON CITY UNIVERSITY

